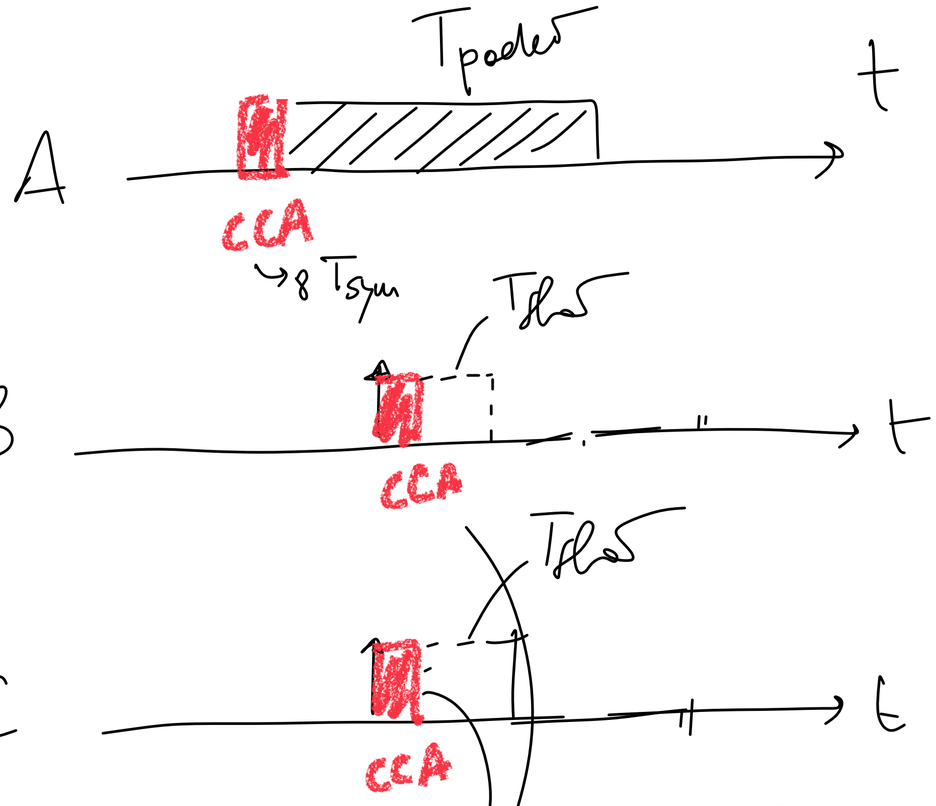
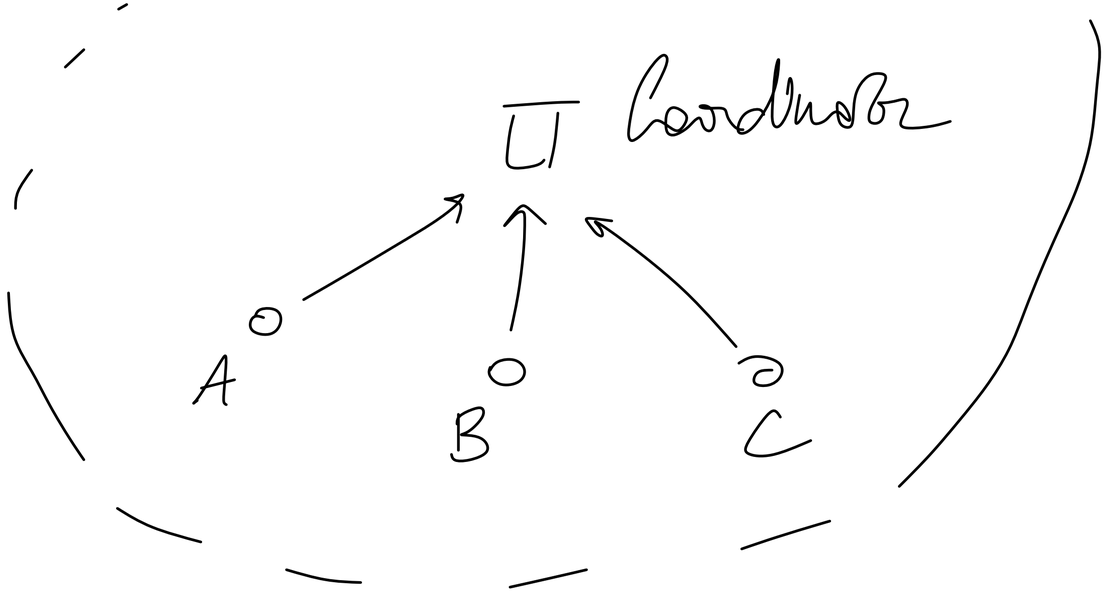
IoT - Lecture 9 - Mar. 25, 2026

802.15.4



B and C perceive that the channel is busy.

1



CSMA/CA

Before sending a packet, every node senses the channel

$BE_1 = 1$

1) B, C → Wait for a number of T slots randomly selected in the set $\{0, 2^{BE_1} - 1\} = \{0, 1\}$ — CA Mechanism

Denote as $n_{slot}^{(B)}$ and $n_{slot}^{(C)}$ the number of slots

$n_{slot}^{(B)} \in \{0, 1\}$ $n_{slot}^{(C)} \in \{0, 1\}$

After this number of T_{slot} intervals have elapsed → they do CCA again

2) If the channel is sensed busy again

$$BE_2 = BE_1 + 1 = 1 + 1 = 2$$

$$\{0, \dots, 2^{BE_2} - 1\} = \{0, 1, 2, 3\}$$

A node selects randomly a number in this set and waits for $n^{(B)}_{slot} \cdot T_{slot}$ (or $n^{(C)}_{slot} \cdot T_{slot}$).

3) If the channel is sensed busy $\rightarrow$ $BE_3 = BE_2 + 1 = 3$

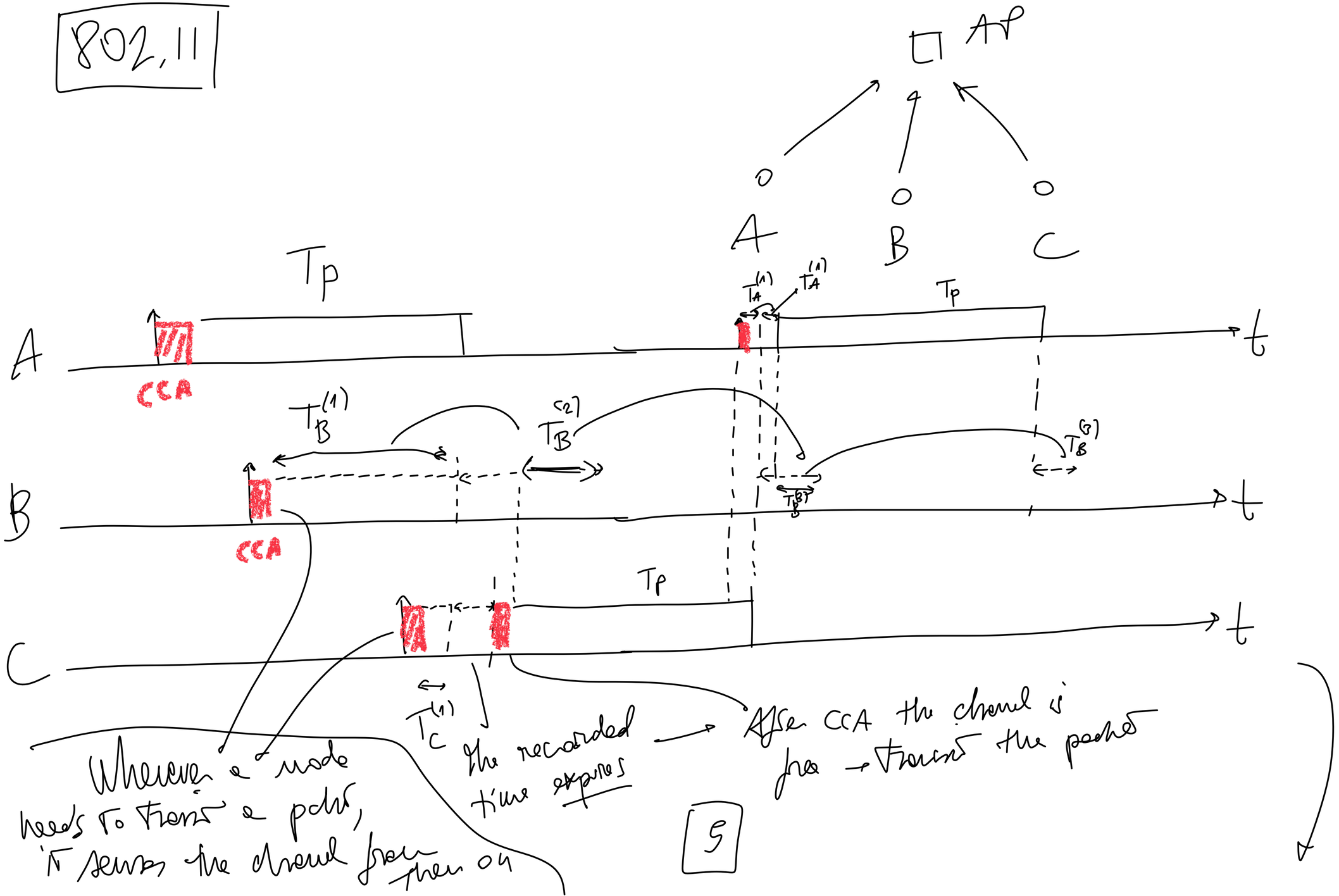
$\{0, \dots, 2^{BE_3} - 1\} = \{0, 1, \dots, 7\}$

3

↳ Why is this a Collision Avoidance (CA) mechanism?

Intuitively, if the BE increases, then the set of possible delays increases ⇒ more unlikely that two nodes select the same delay and start transmitting colliding.

802.11


AP
A
B
C
Tp
CCA
t
Tp
Tp
Whenever a node needs to transmit a packet, it senses the channel for then on
The recorded time expires
After CCA the channel is free → transmit the packet
5

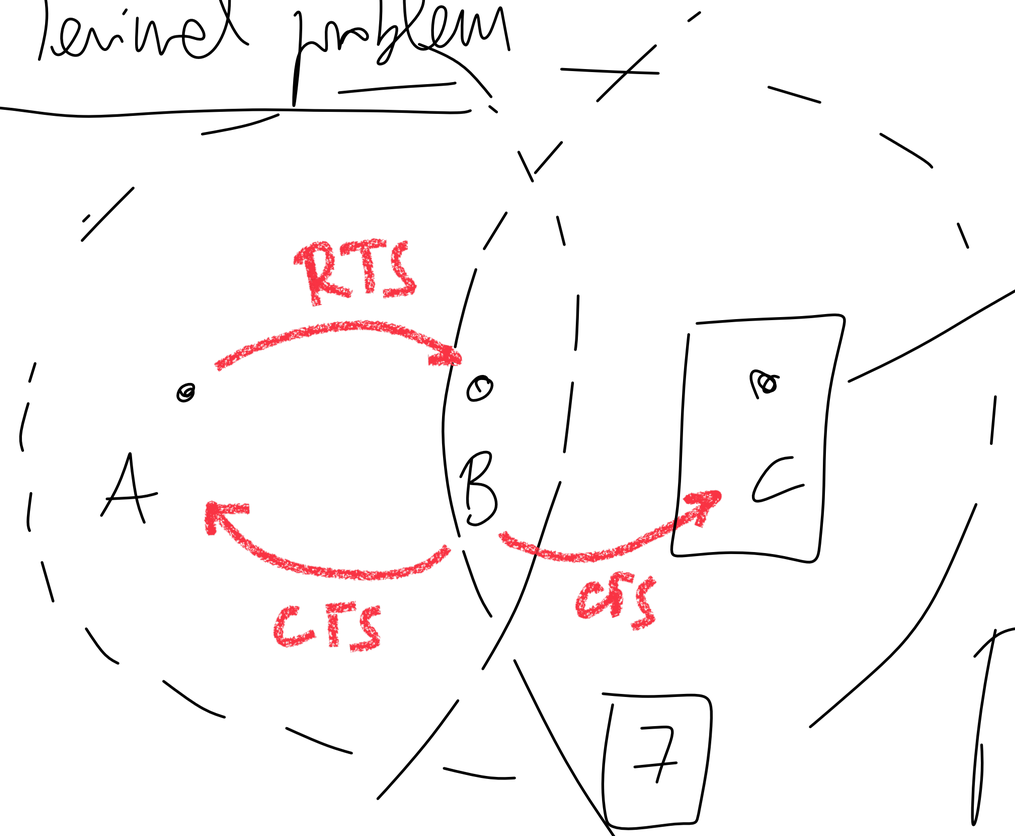
Expert CA mechanism because it is in form (B knows the first one to generate a packet after A stopped transmitting and is the last one to finish it)

RTS & CTS → Control packets

Ready To send

Clear To send

Hidden Terminal problem

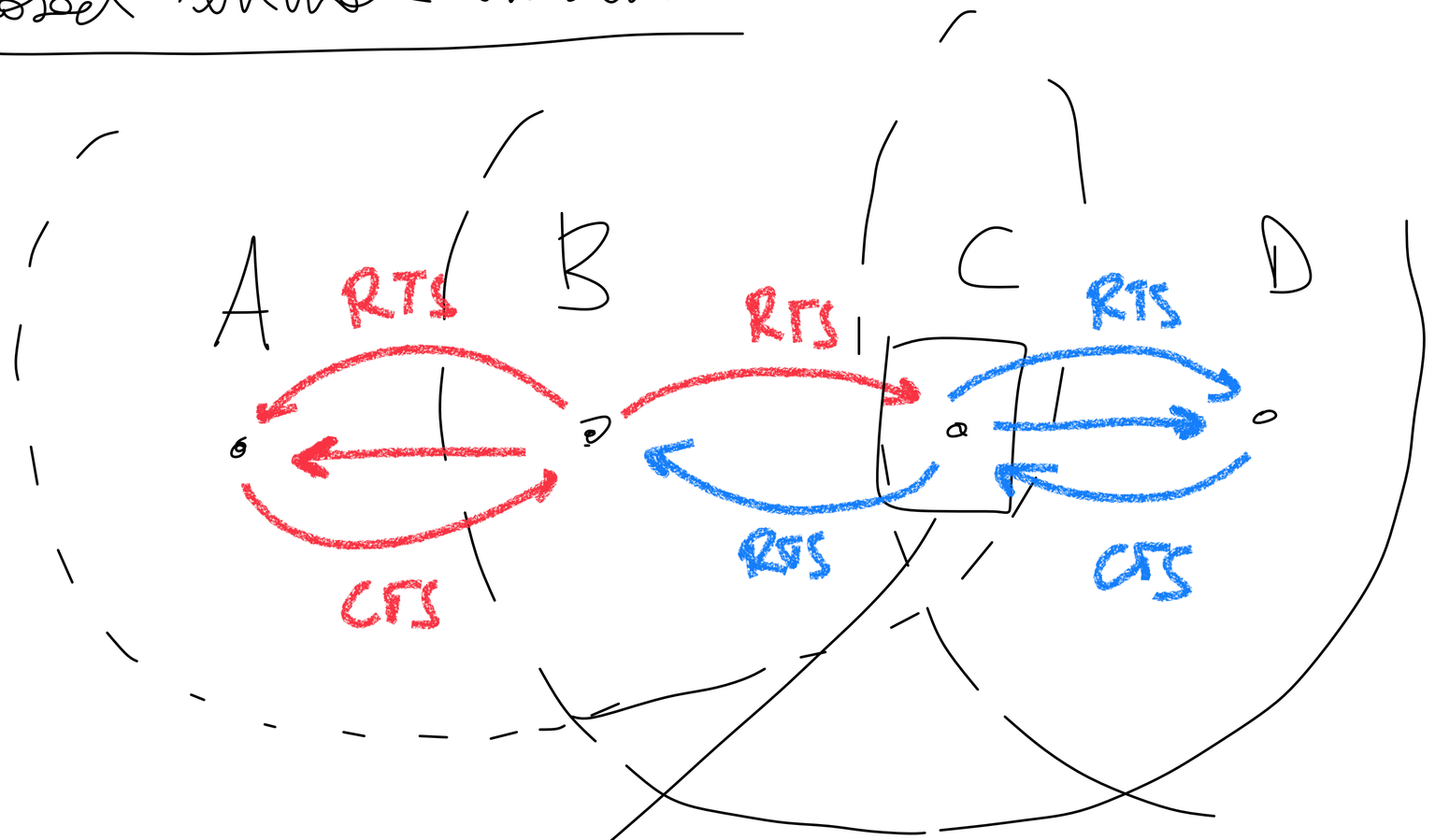


C receives a CTS (addressed To A)

did not receive the RTS

Concludes that it is within the transmission range of receiver

Exposed Terminal Problem



C → receive the RTS
→ did not receive the CTS
⇒ C concludes that it is within the transmission range of the transmitter, but out of the receiver

Can start transmitting to D !